

EFY Note

The source code of this project is available for free download at source.efymag.com

ously until any button (SS1 through SS8 or ES1 through ES5) is pressed by the user. Also, the new-message LED3 glows and a beep is heard whenever a new message is received. Other sets, which do not match the ID_NO, discard the message.

The username and time (in HH:MM format) available in Module1 are also displayed in first row of the LCD in Module2 and the actual received message is displayed in second row of the LCD. The username of Module1 will be saved in the EEPROM (of microcontroller) of Module2. Up to ten such received messages can be saved in each module. But only the latest message received by each module is displayed on the LCD.

To view all the received and stored messages, press ES3 (for next received

message) and ES4 (for previous received message) buttons. To delete any message after reading, press ES2 button. On pressing ES3 and ES4 buttons, a small beep is heard if end of messages is reached.

User definitions

The following usernames and messages are defined in the code (GroupMessenger.C). You can change them as per your requirement. Then compile the code using AVR Studio 4 (or above) and write the new *GroupMessenger.hex* file to the microcontroller.

Usernames. In the below given definition, each username with ten characters (use space if less than ten characters) can be used.

```
char USER_NAMES[8][11] = { "UserName01",
"UserName02",
"UserName03", "UserName04", "UserName05",
"UserName06", "UserName07", "UserName08"
};
```

Messages. The following messages are defined in the code. The length of a message should not exceed sixteen characters, otherwise the message will

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be truncated in the receiver's LCD.

```
char *USER_MESSAGES[] = { "Send e-mail",
"Attend Meeting", "Contact Me", "Call
Back", "Job Status ?", "OK / Done", "I am
Leaving", "On Leave Tomorrow", "Bring File",
"Contact Client", "Contact Head", "Get
Application", "Ready.", "In Progress...",
"Waiting...", "Logged In";
```

Notes.

1. The microcontroller may go to endless loop for reception of acknowledgment from RTC chip on any error or damage, or missing of DS1307.

2. The message sent will never be received by receiver module if it is powered off.

3. A rechargeable 11.1V, 2200mAh Li-ion battery may be used as power supply.

4. The circuit was tested for reception up to 50 metres through walls and slabs and obtained satisfactory results. **EFY**



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